

Jeril Kuriakose

PRINCIPAL DATA SCIENTIST — GENERATIVE AI / LLMS

Riyadh, Saudi Arabia · jerilkuriakose10@gmail.com · +966 50 951 4280

jerilkuriakose.github.io · linkedin.com/in/jerilkuriakose · github.com/jerilkuriakose · scholar.google.com/citations?user=qRQImIoAAAAJ

Summary

Principal Data Scientist with deep expertise in LLMs, NLP, and agentic AI systems. Leads end-to-end development of high-impact platforms from data engineering and training to MLOps, inference optimization, and secure deployment at scale. Proven record of measurable business outcomes (productivity savings, accuracy gains, and risk reduction) and hands-on leadership of cross-functional data teams.

Recent focus: Arabic LLMs (ALLaM), agent orchestration, curriculum SFT, DPO, RL-SFT hybrid training, and large-scale inference on Kubernetes with vLLM/TGI/Triton, Ray, and Azure.

Technical Skills

ML/AI: LLMs, NLP, Agentic systems, RAG, Time-series

Languages & Frameworks: Python, PyTorch, Transformers, TRL, Data-Juicer, vLLM, Ray, LangGraph, Dify

MLOps/Infra & Data: Kubernetes, Azure, MLflow, ClickHouse, Qdrant (Vector DB)

AI-Augmented Dev: Claude Code, OpenCode, Codex (custom skills & agents for workflow automation)

Experience

Principal Data Scientist (Gen AI)

Saudi Data & AI Authority (SDAIA) · Riyadh, Saudi Arabia · Jan. 2024 – Present

- Lead data processing and inference optimization for ALLaM (Arabic LLM); managed training/inference stacks with PyTorch, Transformers, TRL, vLLM/TGI/Triton, Ray, Kubernetes, Azure.
- Designed curriculum SFT pipeline for agentic capabilities; improved model convergence by ~35%.
- Processed 50TB RedPajama-Data-v2 via Data-Juicer; implemented LLaMA-2 pretraining strategies (from-scratch and continual).
- Deployed/optimized ALLaM 7B/13B/70B on Kubernetes; identified optimal inference parameters across use cases.
- Implemented agentic RAG and multi-agent planning/orchestration using LangChain, LangGraph, and Dify.
- Designed AI-augmented development workflows using Claude Code, OpenCode, and Codex; authored custom skills and agents to automate engineering tasks and accelerate delivery.
- Built the post-training stack for agentic capabilities: interleaved RL with periodic SFT injections, iterative DPO with uncertainty-aware preference collection, self-optimized fine-tuning (SOFT) via self-distillation, and gradient separation across RL/SFT objectives.

Senior Data Scientist

Mizuho Bank · Singapore · Jul. 2019 – Dec. 2023

- Phoenix: generic NLP IE platform using BERT + hybrid CNN + Bloom embeddings; ~90% accuracy; saved ~100,000 person-hours/year.
- PIGEON: document classification with RoBERTa and summarization (GPT-2/GPT Neo); achieved 97% accuracy; full API productionization.
- SWAN: AML name check reduced false positives by 20%; vector DB for embeddings; graph neural networks for anomaly detection.
- HAWK: hybrid encoder-decoder LSTM for daily/weekly/monthly forecasts; saved ~3,000 person-hours/year; external data ingestion + ELK.
- Introduced MLflow for org-wide model tracking; implemented ML monitoring and performance validation in production.

Senior Data Scientist

Baker Hughes (GE Company) · Kochi, India · Jan. 2019 – Jun. 2019

- Predicted stuck-pipe events with feature-engineered ensemble models; identified key drivers and timing of incidents.
- Analyzed non-productive time (NPT) and drill-bit wear; delivered actionable insights to reduce rig downtime.
- Drove awareness of applied data science among SMEs through targeted training.

Senior AI Engineer

Innovation Incubator · Thiruvananthapuram, India · Feb. 2018 – Jan. 2019

- Generic keyword extraction across PDFs/TXT/Excel/HTML via ensemble ML and feature engineering; production APIs.
- Customer segmentation and claim-settlement prediction; improved healthcare RCM by ~18%.
- Owned end-to-end ML pipelines and backend services; collaborated with customers to translate business to ML problems.

Senior ML/AI Developer

Raw Data Technologies · Kochi, India · Apr. 2017 – Feb. 2018

- EMI defaulter prediction with ensemble models; extensive feature engineering; deployed APIs and data pipelines.
- Performance analysis of internal vs external collection agents using Kaplan–Meier estimator and cumulative incidence.

Earlier experience (2012–2017): Assistant Professor, Developer, and Data Analyst roles at St. John College of Engineering and Technology, Manipal University Jaipur, and Kavary College of Engineering (India) — teaching ML/Data Mining, university ERP development (Django), and student-outcome predictive analytics.

Publications

ALLaM: Large Language Models for Arabic and English

Bari M. S., Kuriakose J., et al. · The International Conference on Learning Representations (ICLR) · 2025

A review of deep learning-based approaches for detection and diagnosis of diverse classes of drugs

Kuriakose J., et al. · Archives of Computational Methods in Engineering (Springer) · 2023

EMBN-MANET: Eliminating Malicious Beacon Nodes in UWB-based Mobile Ad-Hoc Networks

Kuriakose J., Joshi S., Bairwa A. K. · Ad Hoc Networks (Elsevier) · 2022

Secure Multipoint Relay Node Selection in Mobile Ad Hoc Networks

Kuriakose J., Amruth V., Raju R. V. · Communications in Computer and Information Science (Springer) · 2015

A Review on Mobile Sensor Localization

Kuriakose J., et al. · Security in Computing and Communications (Springer) · 2014

Selected publications; 35+ total — see Google Scholar

Education

Ph.D. in Computer Engineering

Manipal University Jaipur, School of Computing and IT · Jaipur, India · Jul. 2013 – Dec. 2019

M.Tech. in Information Technology

ISIM, University of Mysore · Mysuru, India · Jul. 2012

B.Tech. in Information Technology

Jeppiaar Engineering College (Anna University) · Chennai, India · May 2010

Awards

Project of the Year · Mizuho Bank · Jul. 2022

Employee of the Month · Mizuho Bank · Oct. 2019, May 2021, Jul. 2021

Honourable Award – AML Project · Mizuho Bank · Feb. 2020

Honourable Mention – Phoenix Project · Mizuho Bank · Dec. 2020

Languages

English, Hindi, Tamil, Malayalam, Arabic